Steady Two-Dimensional Heat Conduction in a Rectangular Plate

Comparison of point-Jacobi and line-TDMA/SOR iterative techniques

1. Aim

To solve the steady two-dimensional heat-conduction problem governed by Laplace's equation and compare the performance of point-Jacobi iteration with a line-implicit TDMA/SOR approach. The numerical temperature fields are assessed against the analytical solution, with attention to convergence and grid resolution.

2. Physical Model and Parameters

The problem considers a rectangular plate with a prescribed temperature distribution along its left boundary. The remaining three boundaries are maintained at zero temperature. The peak temperature applied at the heated boundary is 100, and the resulting steady temperature field is computed on multiple meshes.

Parameter	Specification
Governing equation	Two-dimensional Laplace equation
Left boundary	T(0,y) = T_L sin(pi y/H)
Right, top and bottom boundaries	T = 0
Peak source temperature	100
Maximum iterations	10,000
Convergence tolerance	1 x 10^(-9)
Meshes reported	81 x 61; 30 x 15; 51 x 51

3. Governing Equation and Boundary Conditions

For steady conduction with no internal heat generation and constant thermal properties, the temperature field satisfies the two-dimensional Laplace equation. The left edge is assigned a sinusoidal temperature profile, T(0,y) = T_L sin(pi y/H). The right, upper and lower edges are held at zero temperature.

4. Numerical Methods

Second-order central finite differences are used to discretize the governing equation. The point-Jacobi method updates each interior node using values from the previous iteration. The line-TDMA method solves along grid lines using a tridiagonal system, with SOR relaxation used to accelerate convergence. Both approaches are iterated until the residual reaches the specified tolerance or the maximum iteration limit is reached.

5. Quantities Compared

The methods are compared using the iteration history of the L-infinity residual, final residual, iteration count, relative L2 error and horizontal centreline temperature profile. The temperature contours are also inspected to assess the spatial distribution of heat across the plate.

6. Results and Convergence Behaviour

The source report contains results for three mesh configurations. The numerical values available in the report are summarized below; entries described as requiring actual output have not been estimated.

Mesh	Method / quantity	Reported result
30 x 15	Jacobi iterations	About 1,800
30 x 15	Line-TDMA iterations	About 200
81 x 61	Jacobi iterations shown	10,000
81 x 61	Jacobi final L-infinity residual	About 2 x 10^(-6)
81 x 61	Line-TDMA iterations	About 800–900
81 x 61	Line-TDMA final residual	Below 10^(-9)
51 x 51	Iteration counts and final residuals	Actual solver output required

7. Discussion

For the 30 x 15 mesh, the line-TDMA method reaches the specified tolerance in approximately 200 iterations, compared with roughly 1,800 iterations for Jacobi. This indicates a substantially faster convergence rate for the line-implicit approach on the coarse grid.

On the 81 x 61 mesh, the line-TDMA method reaches a residual below 10^(-9) in approximately 800–900 iterations. The Jacobi calculation is shown running to 10,000 iterations, at which point its residual is approximately 2 x 10^(-6), still above the specified tolerance. The convergence plot on source page 2 shows the line-TDMA residual falling much more rapidly than the Jacobi residual.

The centreline plots on source pages 2 and 4 show close overlap between the Jacobi, line-TDMA and analytical temperature profiles. The temperature-field plots on source pages 3 and 5–6 show the highest temperatures near the centre of the heated left boundary, with temperature decreasing toward the zero-temperature edges.

The additional 51 x 51 mesh is included to examine the effect of grid resolution. However, the source results table identifies its actual iteration counts and final residuals as requiring solver output. Therefore, its convergence performance cannot be quantified from the supplied report.

8. Conclusion

The steady two-dimensional plate-conduction problem was solved using point-Jacobi and line-TDMA/SOR methods on several grid configurations. The centreline temperature profiles are reported to agree closely with the analytical solution, and the contour plots show the expected decrease in temperature away from the heated boundary.

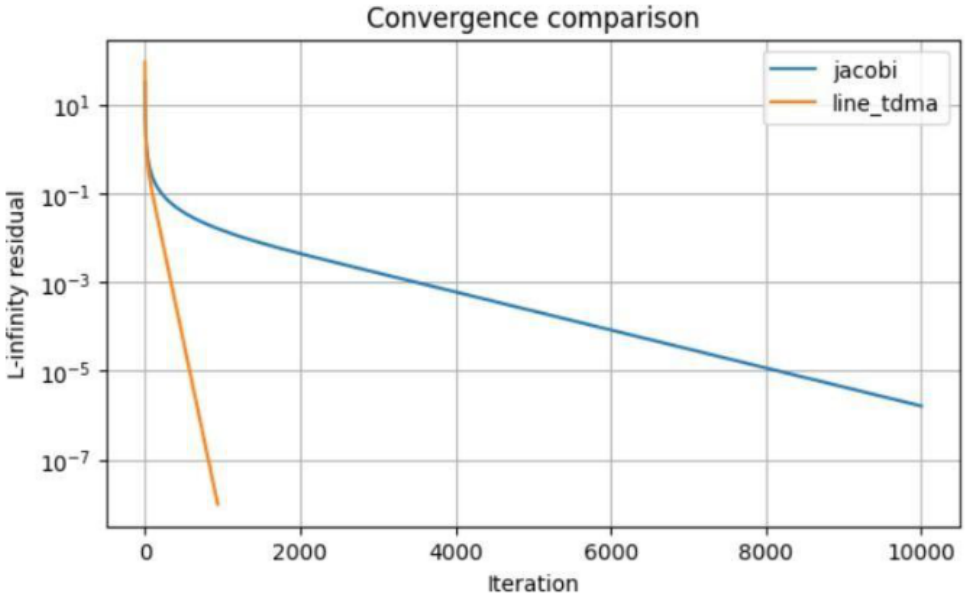
The available convergence results show that line-TDMA is considerably faster than Jacobi. On the 30 x 15 mesh, the reported iteration counts are approximately 200 for line-TDMA and 1,800 for Jacobi. On the 81 x 61 mesh, line-TDMA reaches the 10^(-9) tolerance in about 800–900 iterations, whereas Jacobi remains at a residual of approximately 2 x 10^(-6) after 10,000 iterations.

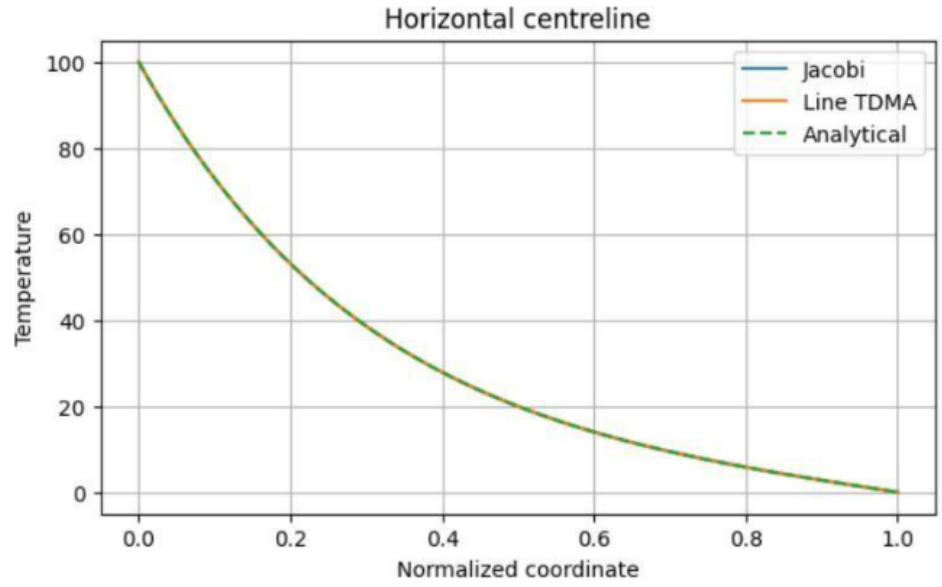
The 51 x 51 case provides an additional grid comparison, but its final iteration counts and residuals are not available in the supplied results. These values are needed before making a quantitative comparison of its convergence rate and accuracy. Based on the reported cases, line-TDMA/SOR is the more efficient approach for this problem.

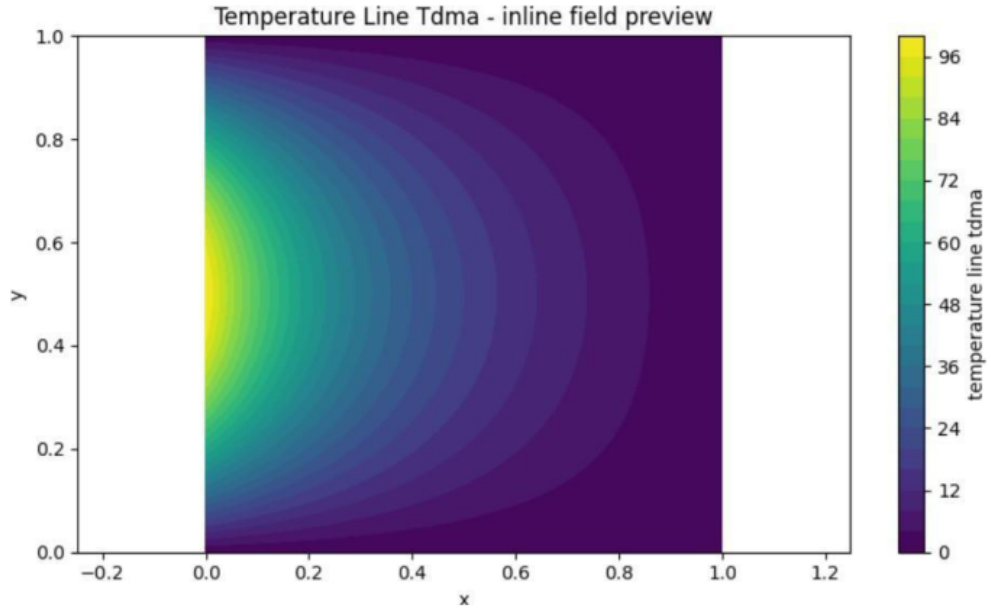
9. Original Simulation Figures

The original residual histories, centreline comparisons and temperature-field contours are reproduced in the following pages.

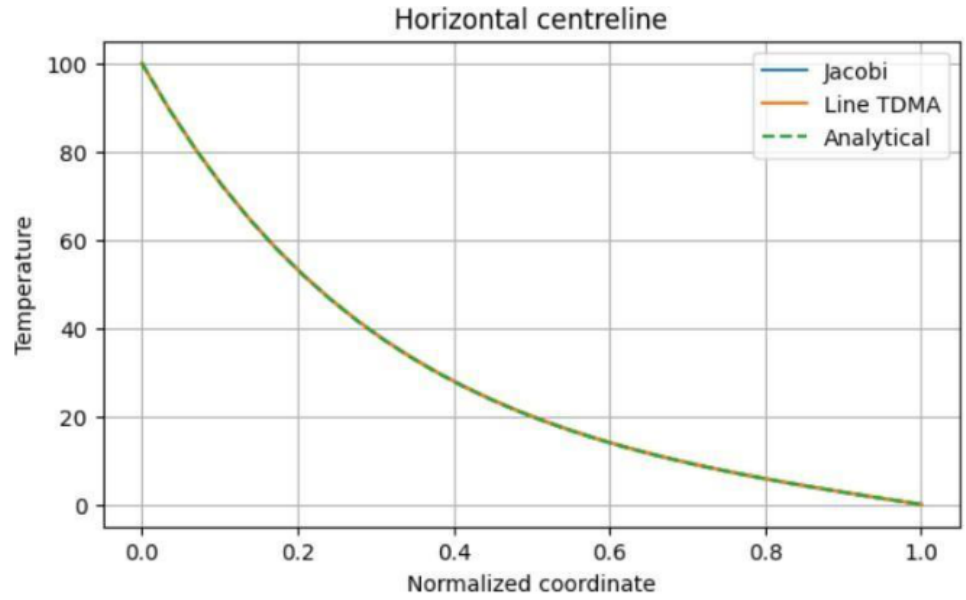
For nx=81 and ny=61

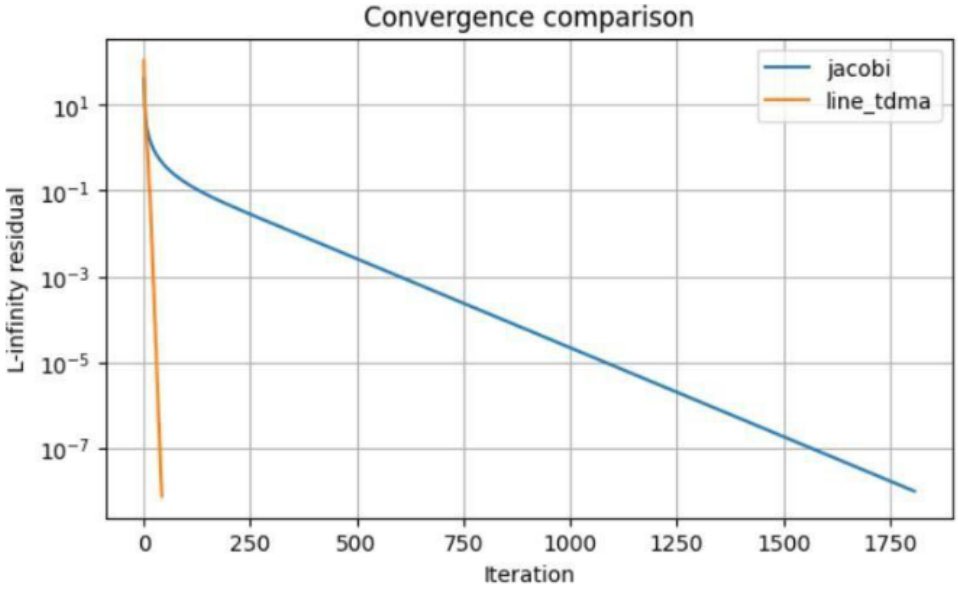


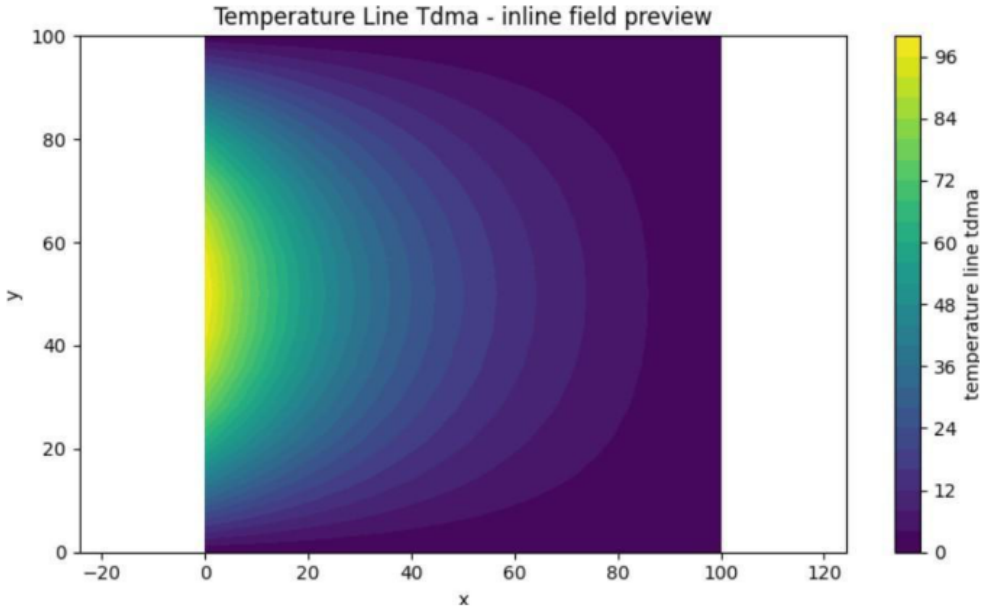




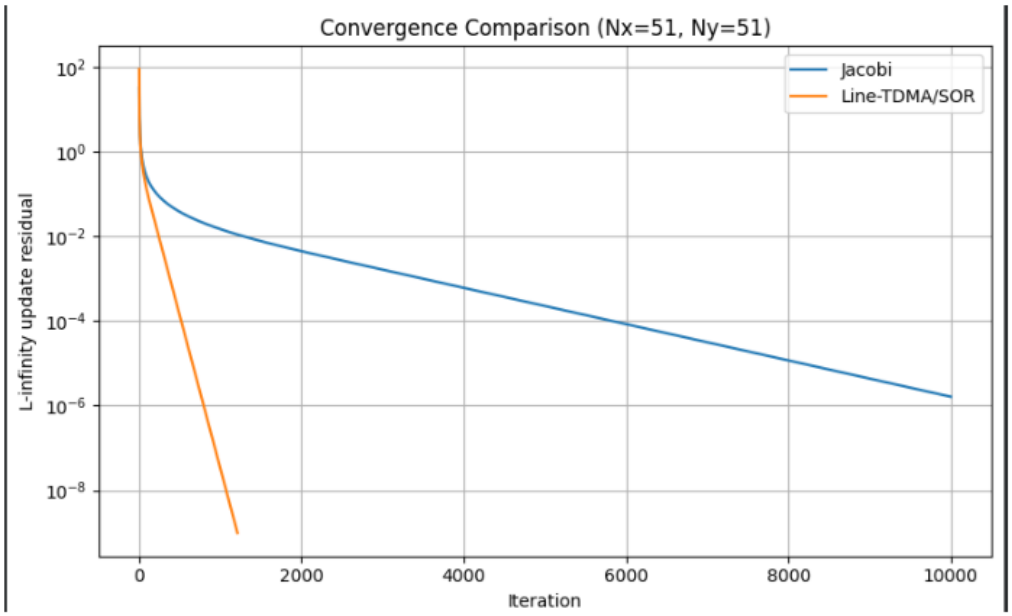
For nx=30 and ny=15

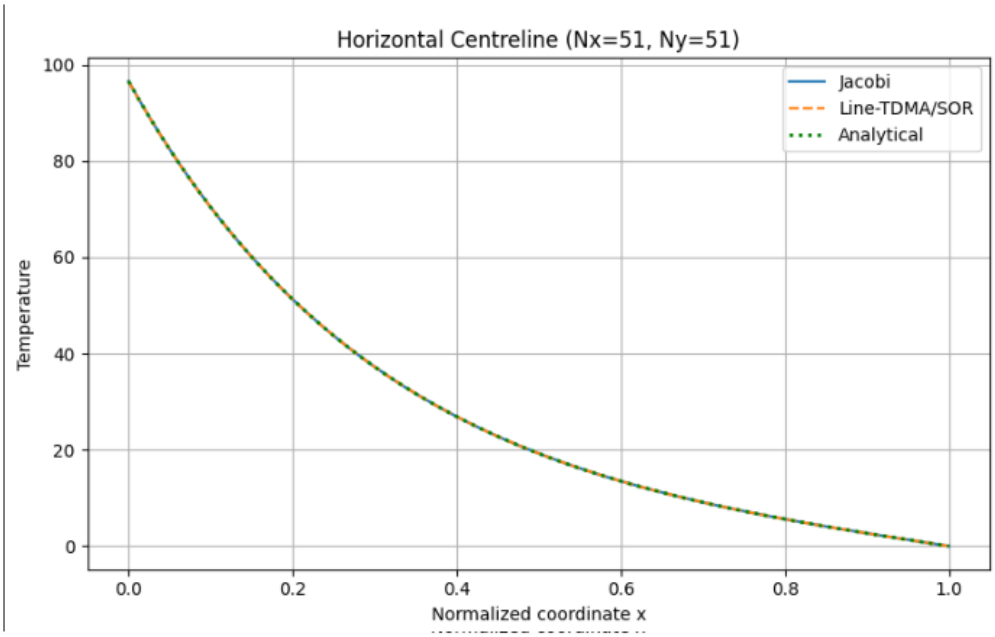

Horizontal centreline
Jacobi
Line TDMA
Analytical
Temperature
Normalized coordinate
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0

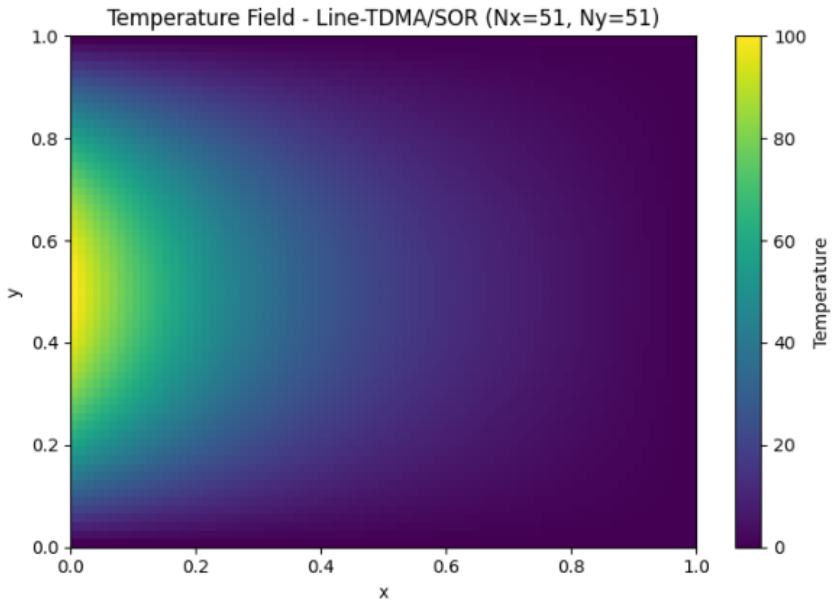

Convergence comparison
jacobi
line_tdma
L-infinity residual
Iteration
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
0
250
500
750
1000
1250
1500
1750


Temperature Line Tdma - inline field preview
y
x
temperature line tdma
100
80
60
40
20
0
−20
0
20
40
60
80
100
120
96
84
72
60
48
36
24
12
0

For 51 × 51


Convergence Comparison (Nx=51, Ny=51)
L-infinity update residual
Iteration
Jacobi
Line-TDMA/SOR
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
2000
4000
6000
8000
10000


Horizontal Centreline (Nx=51, Ny=51)
Jacobi
Line-TDMA/SOR
Analytical
Temperature
Normalized coordinate x
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0


Temperature Field - Line-TDMA/SOR (Nx=51, Ny=51)
y
x
Temperature
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100

Results:

Parameter	**Fine Grid (81 × 61)**	**Additional Grid (51 × 51)**
Grid size, (N_x \times N_y)	81 × 61	51 × 51
Source temperature	100	100
Maximum iterations allowed	10,000	10,000
Convergence tolerance	(10^{-9})	(10^{-9})
Jacobi iterations shown	10,000	Actual output required
Line-TDMA iterations to reach tolerance	800–900	Actual output required
Jacobi final (L_\infty) residual	(2 \times 10^{-6})	Actual output required
Line-TDMA final (L_\infty) residual	Below (10^{-9})	Actual output required
Centreline profile agreement	Very good	Verify against analytical solution
Relative convergence speed	Slower	Depends on actual iteration counts

Conclusion:

Conclusion

The two-dimensional steady heat conduction problem was solved using the Jacobi and line-TDMA iterative methods on grids of 81 × 61, 30 × 15, and an additional grid of 51 × 51. Both methods produced temperature distributions that closely matched the analytical centreline solution, with a maximum temperature of approximately 100 at the midpoint of the heated boundary and a gradual decrease toward the zero-temperature boundaries.

The convergence results showed that line-TDMA was substantially more efficient than Jacobi. On the 30 × 15 coarse grid, Jacobi required approximately 1,800 iterations to reach the specified tolerance of (10^{-9}), whereas line-TDMA required approximately 200 iterations. On the 81 × 61 fine grid, line-TDMA reached the tolerance within approximately 800–900 iterations, while Jacobi remained at a residual of approximately (2 \times 10^{-6}) after 10,000 iterations.

An additional simulation was performed using a 51 × 51 grid to investigate the effect of grid resolution on the temperature distribution and convergence behaviour. This grid provides an intermediate comparison with the original fine grid. Its actual iteration counts and final residuals should be used to quantify the convergence performance and determine whether the specified tolerance was achieved.

Overall, the results demonstrate that grid resolution affects the computational effort required for convergence, particularly for point-Jacobi iteration. The line-TDMA method handles this increase more efficiently because of its line-implicit solution procedure. Therefore, line-TDMA is the preferred method for this problem when faster convergence and satisfaction of a strict residual tolerance are required. The additional 51 × 51 results provide further information for evaluating the relationship between grid resolution, convergence speed, and solution accuracy.